

MQ-135 • DHT11 • LDR MODULE — ESP8266 NODEMCU → BLYNK IOT → ML CLASSIFIERS

SMART INDOOR AIR QUALITY MONITORING

An Embedded IoT System with Machine-Learning-Based Air Quality Classification

Sensor data acquisition → Real-time dashboard → Multi-model training & evaluation of Air Quality

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From Sensors to a Trained Air-Quality Classifier

A three-stage pipeline: embedded sensing, live IoT visualization, and offline machine-learning analysis that predicts Air Quality.

01

Embedded Sensing

ESP8266 NodeMCU reads MQ-135 (gas), DHT11 (temp/humidity) and an LDR module every 10 s, then classifies Air Quality and flags gas anomalies on-device.

02

IoT Dashboard + Logging

Readings are streamed live to Blynk for monitoring, and appended to a Google Sheet (exported as CSV) for long-term storage.

03

Machine Learning

The logged CSV is loaded in Colab, cleaned & scaled, then used to train four classifiers — Logistic Regression, ANN, Random Forest, KNN — to predict Air Quality (Moderate / Poor).

Sensor Hardware & On-Device Classification Logic

MQ-135 Gas Sensor

Analog — A0

Raw ADC gas concentration reading (0–1023); primary driver of Air Quality

DHT11 Sensor

Digital — D4

Ambient temperature (°C) and relative humidity (%)

LDR Module

Digital — D5

Light state, used as a proxy for room occupancy

ESP8266 NodeMCU

Wi-Fi MCU

Samples every 10 s; pushes data to Blynk + Google Sheets

Air Quality Classification Rule

Derived on-device from the MQ-135 raw ADC reading:

< 300 **Good**

300 – 599 **Moderate**

≥ 600 **Poor**

Model target used for training: the ML pipeline predicts this same 'Air quality' label from the logged sensor history.

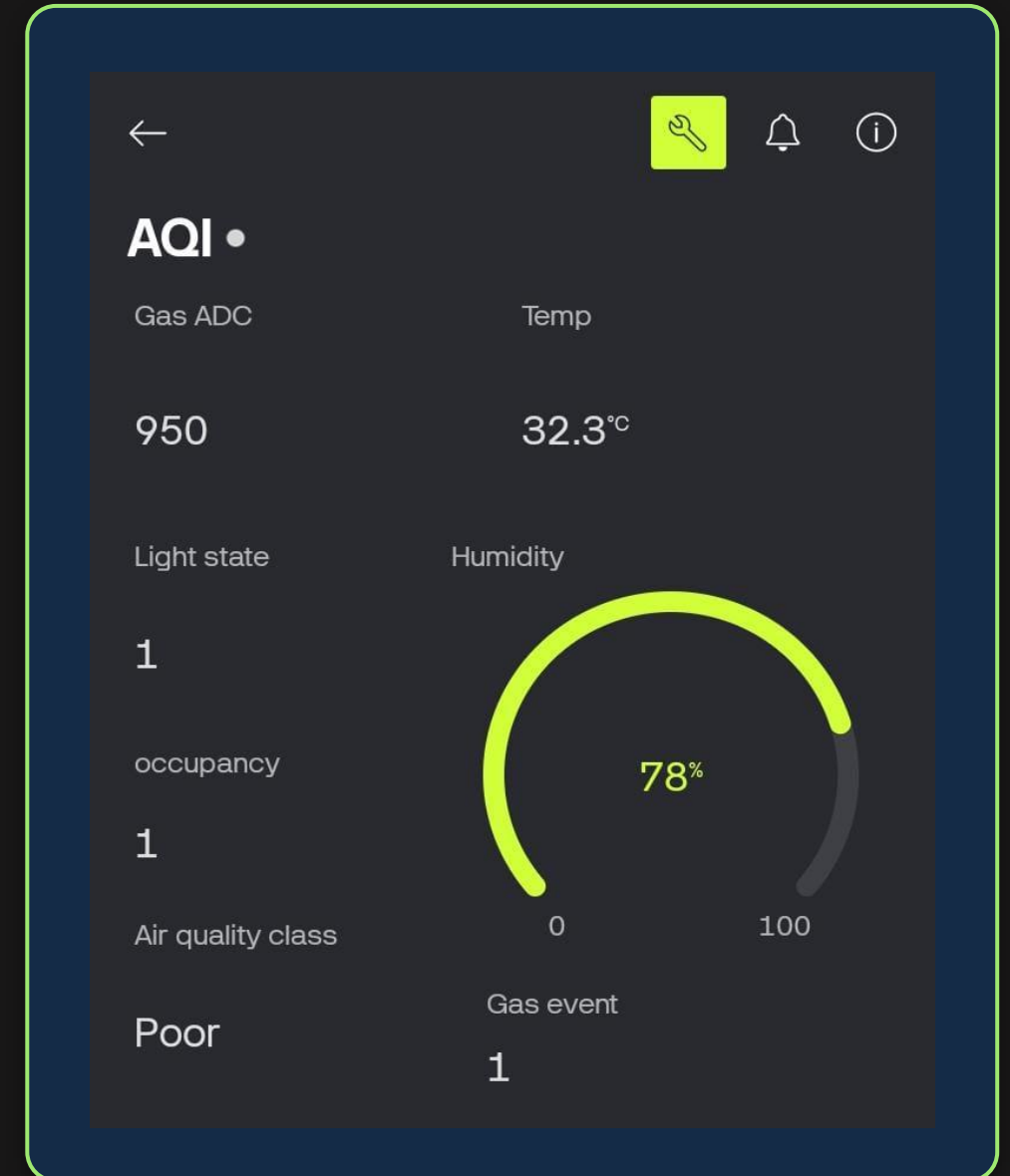
Gas Anomaly Flag

MQ-135 raw value > 600 → Anomaly = 1 (secondary, on-device safety flag alongside Air Quality)

Real-Time Parameters on Blynk IoT

Every 10 seconds, the NodeMCU writes seven parameters to Blynk virtual pins for live, remote monitoring.

- **Gas ADC (MQ-135)** — Raw analog gas reading
- **Temperature** — From DHT11, in °C
- **Humidity** — From DHT11, shown on radial gauge
- **Light State** — Digital LDR output (0/1)
- **Occupancy** — Estimated from light state
- **Air Quality Class** — Good / Moderate / Poor
- **Gas Event** — Binary anomaly flag



From Logged Readings to an Air-Quality Training Set

1,499

Logged readings

6

Input features used

90 / 10

Poor vs. Moderate split (%)

Model Input Features

Feature	Source
Temperature (°C)	DHT11
Humidity (%)	DHT11
MQ-135 raw value	MQ-135
Light State	LDR module
Date (encoded)	Timestamp
Time (encoded)	Timestamp

Target label — Air quality (Moderate = 0, Poor = 1 as observed in this logged set)

Preprocessing Pipeline

- 1 Loaded CSV of logged sensor readings into pandas
- 2 Parsed Timestamp → separate Date & Time features (label-encoded)
- 3 Dropped non-predictive / redundant columns (e.g. Occupancy Status)
- 4 Label-encoded the target — Air quality (Moderate / Poor)
- 5 Standardized numeric features with StandardScaler
- 6 Stratified 80 / 20 train–test split for the Air quality target

Four Classifiers Trained to Predict Air Quality

Every model is trained on the same 6-feature, scaled dataset and stratified 80/20 split — 1,199 training samples, 300 test samples.

Logistic Regression

scikit-learn `LogisticRegression`, `max_iter = 1000`. A linear baseline classifier on the scaled features.

Artificial Neural Network

Keras Sequential: `Dense(64, ReLU) → Dense(32, ReLU) → Dense(2, Softmax)`. Adam optimizer, sparse categorical cross-entropy, 50 epochs, batch size 32, 10% validation split.

Random Forest

scikit-learn `RandomForestClassifier` with default ensemble settings — captures non-linear feature interactions.

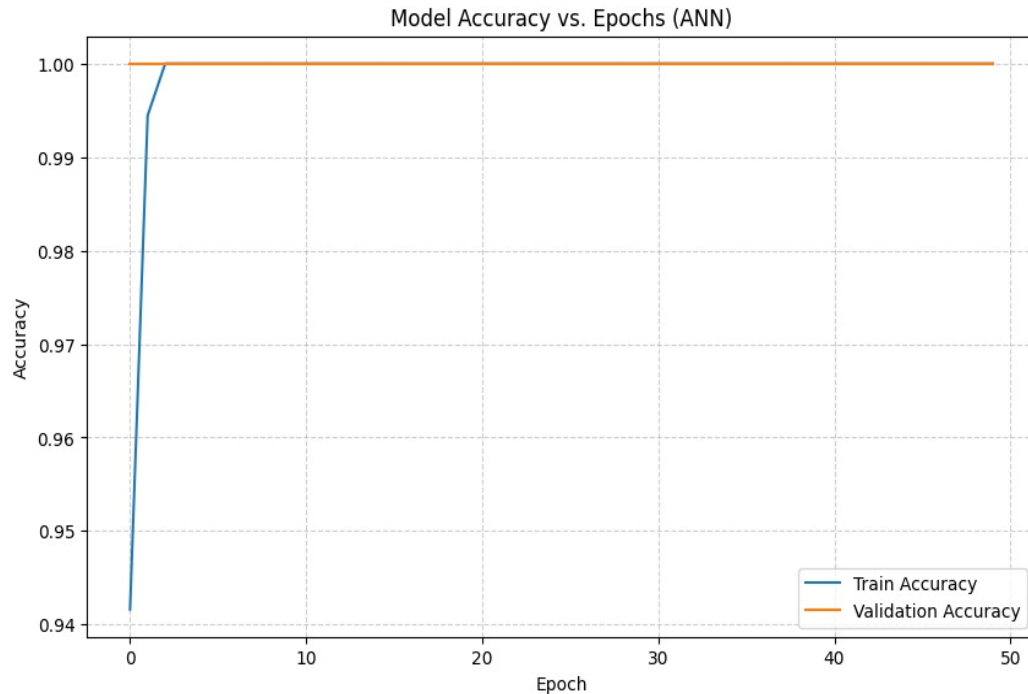
K-Nearest Neighbors

scikit-learn `KNeighborsClassifier`, default `k = 5` — distance-based classification on standardized features.

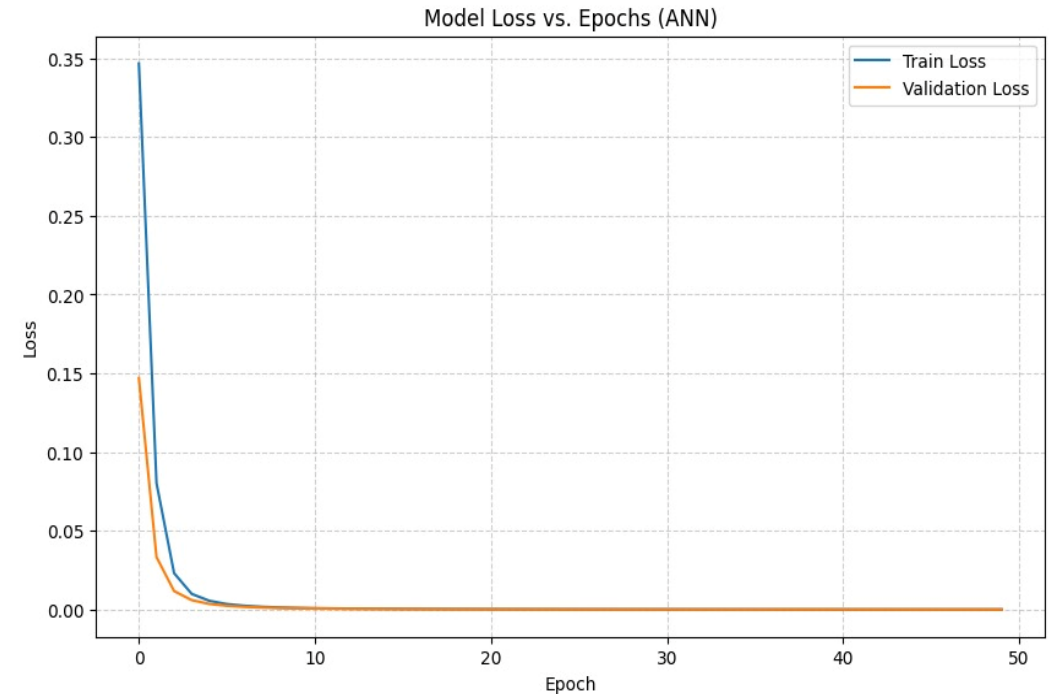
Accuracy & Loss Across 50 Training Epochs

Both training and validation curves converge rapidly, indicating fast, stable learning on the Air Quality classification task.

Model Accuracy vs. Epochs



Model Loss vs. Epochs



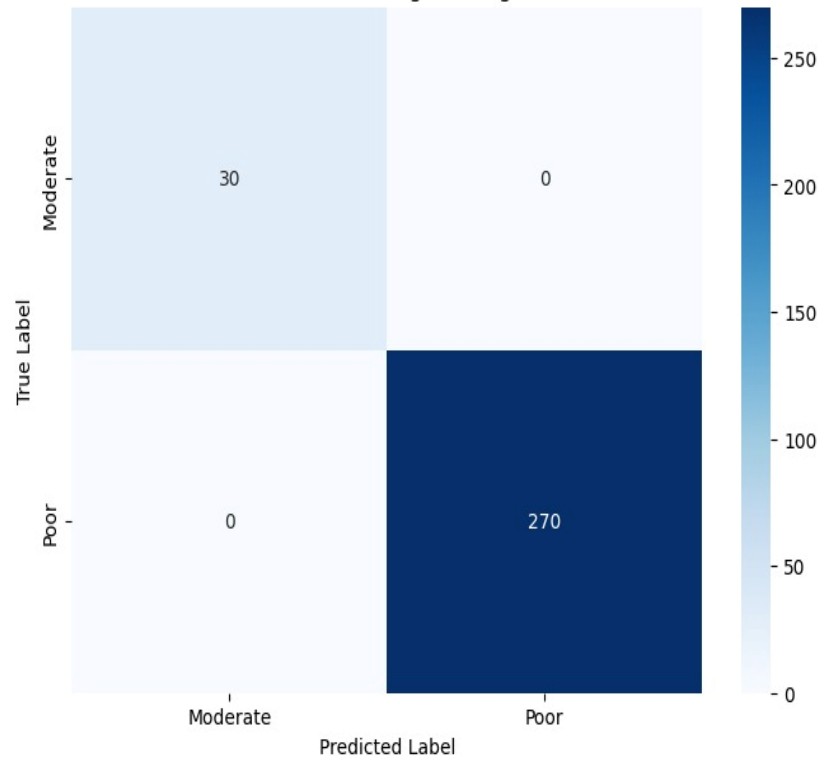
Observation: training and validation accuracy both reach 1.0 within the first few epochs, and loss converges to near zero — consistent with the strong class separability seen across all four models.

Confusion Matrices — Moderate vs. Poor Air Quality

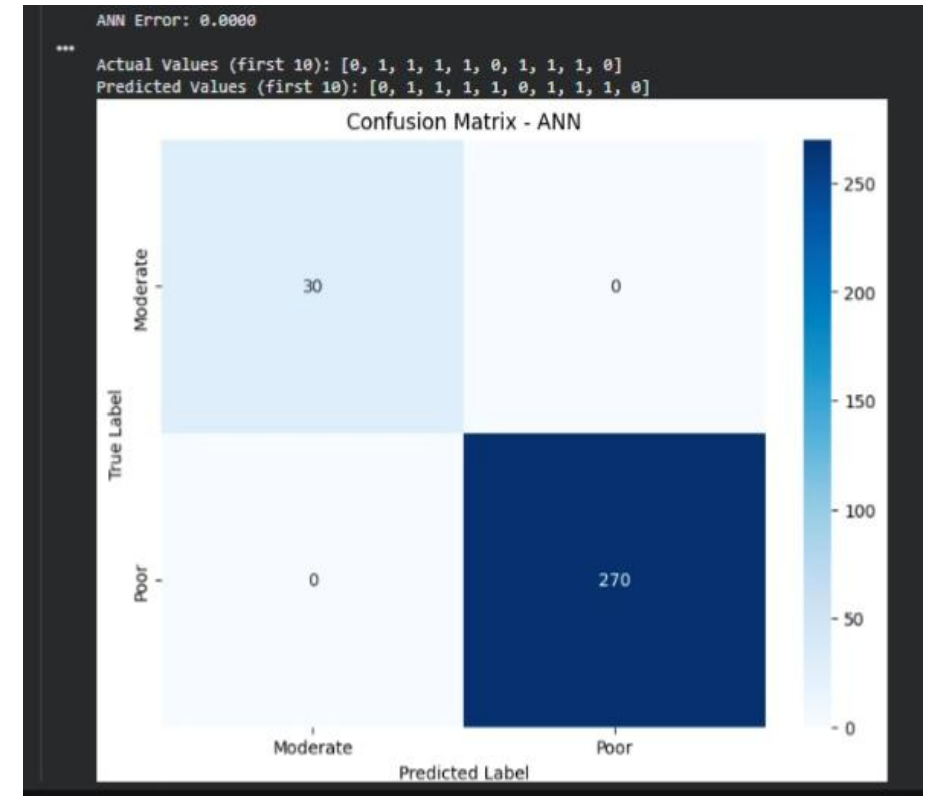
Held-out 300-sample test set (30 Moderate · 270 Poor), evaluated identically across all four classifiers.

Logistic Regression

Confusion Matrix - Logistic Regression



Artificial Neural Network

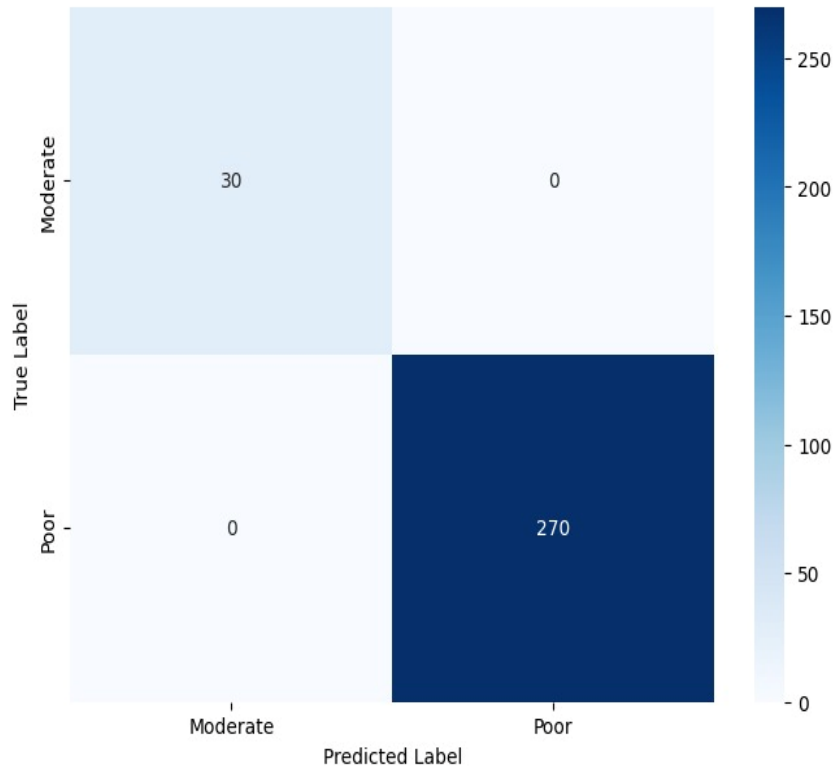


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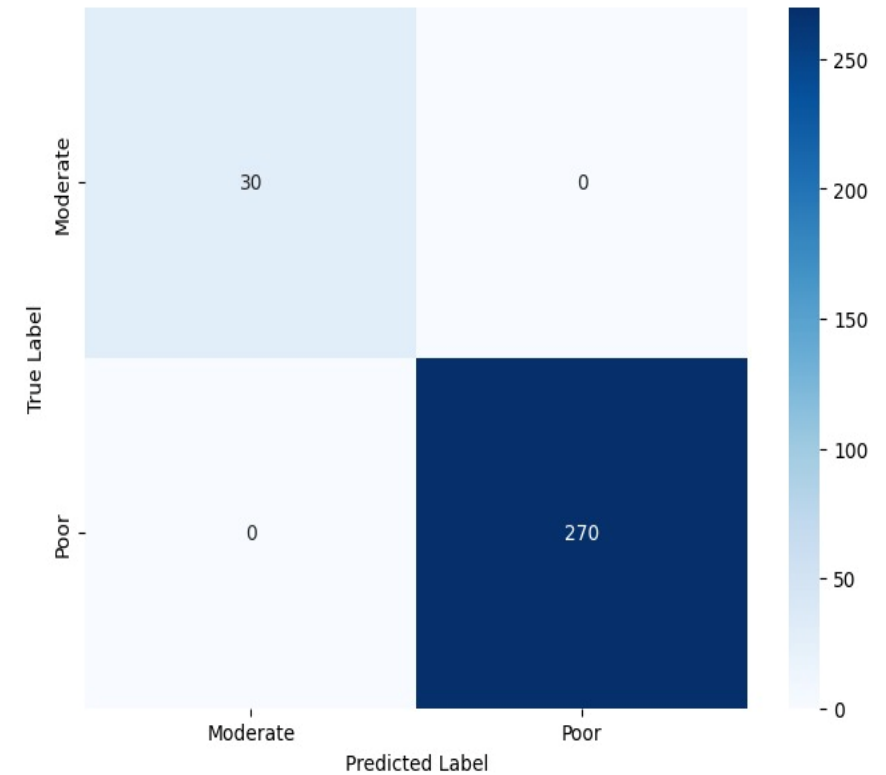
Random Forest

Confusion Matrix - Random Forest



K-Nearest Neighbors

Confusion Matrix - KNN



All Four Models Perfectly Separate Air Quality Classes

Metrics computed with scikit-learn's `accuracy_score`, `precision_score` (weighted) and `f1_score` (weighted) on the 300-sample test set.

Model	Accuracy	Precision	F1 Score	Error
Logistic Regression	1.0	1.0	1.0	0.0
Artificial Neural Network	1.0	1.0	1.0	0.0
Random Forest	1.0	1.0	1.0	0.0
K-Nearest Neighbors	1.0	1.0	1.0	0.0

Reading the results

Perfect scores reflect a small, controlled test setup and a strongly imbalanced dataset (90% Poor readings, captured mostly in one elevated-MQ-135 environment). Results show clean class separability on this data, but broader field testing and a more balanced dataset are recommended before deployment.

Source Output (Colab)

	Model	Accuracy	Precision	F1 Score	Error
0	Logistic Regression	1.0	1.0	1.0	0.0
1	ANN	1.0	1.0	1.0	0.0
2	Random Forest	1.0	1.0	1.0	0.0
3	KNN	1.0	1.0	1.0	0.0

Support: 30 Moderate + 270 Poor = 300 test samples

Takeaways & Next Steps

What we achieved

- End-to-end pipeline built: embedded sensing → live Blynk dashboard → CSV logging → multi-model training
- Low-cost sensor stack (MQ-135, DHT11, LDR) reliably distinguishes Good / Moderate / Poor air quality
- Four models — Logistic Regression, ANN, Random Forest, KNN — each achieved 1.0 accuracy, precision and F1 on Air Quality classification

Where this goes next

- Collect data across more environments to reduce class imbalance and add 'Good' air-quality examples
- Calibrate MQ-135 to real PPM values instead of raw ADC thresholds
- Cross-validate models on a larger, more balanced dataset to stress-test the perfect scores
- Deploy the best-performing model on-device (TinyML) for real-time inference without the cloud round-trip

Thank You